

# Cost Report Guidelines



**SUPRA SAEINDIA 2025**

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# Rules Overview

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## □ Event Objectives:

- To gain **experience** with creating and **maintaining Bill of Material (BOM)**.
- For the participants to **learn** and understand the principles of Design for Manufacture and Assembly, **lean manufacturing and Minimum Constraint Design**.
- No maximum cost –**accuracy is focus**
- Everyone pays the same regardless of where they are in the world
- To require the minimal burden of supporting documentation such as receipts or catalogue pages. However, in order to convey design information to cost judges engineering documentation (**drawings, process descriptions, etc.**) are required.

## □ Event comprised of 3 parts

- Cost Report Submitted prior to the Competition.
- Cost Report/Vehicle Review at Competition
- Cost Report preparedness

# Rules Overview

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- ❑ **At Vehicle Review, the Cost Judges will:**
  - a. Review whether the specification of the vehicle in the Cost Report accurately reflects the vehicle brought to the Competition**
  - b. Review the manufacturing feasibility of the vehicle**
  - c. Assess penalties for missing or incorrect information in the Cost Report compared to the vehicle presented at inspection.**
  
- ❑ **Problems specifically**
  - Could be cost reduction**
  - Could be weight reduction**
  
- ❑ **Usually sent to teams via E-mail 2-3 weeks before the event.**
- ❑ **Teams could use flip charts (optional)**
- ❑ **Must be based on system of your car.**

# Rules Overview

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## □ Scoring

|                                                                            |           |                                                                                                                                                                                                                                                                                       |
|----------------------------------------------------------------------------|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $40 \times \frac{(P_{\max}) - (P_{\text{your}})}{(P_{\max}) - (P_{\min})}$ | 40 Points | Lowest Cost- Each of the participating schools will be ranked by the total adjusted cost from the BOM and given 0-50 based on the formula.                                                                                                                                            |
|                                                                            | 40 Points | Accuracy, Clarity & Event day/visula inspection- The cars will be reviewed for part content, manufacturing feasibility and accuracy of the cost information. Supporting documentation will be assessed based on its quality, accuracy and thoroughness. The scores will be from 0-50. |
|                                                                            | 20 points | Cost report preparedness- It will be judged on the basis of How well teams has prepare their cost report and how well team will be able to explain the content of their reports, Whether report is as per the cost template provided to teams).                                       |
| <b>Total</b>                                                               |           | <b>100 Points</b>                                                                                                                                                                                                                                                                     |

# Rules Overview

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## □ Penalty Methods

### □ Method A –

–Fixed Point Deductions

–Most errors will be penalized using Method A

–Missing/inaccurate material, process, fastener.....1 pt.

–Missing/inaccurate part..... 3 pt.

–Missing/inaccurate assembly..... 5 pt.

### □ Examples-

–Five M6 fasteners listed, six used – 1 pt.

–Three kilograms of steel listed, 4.4 used – 1 pt.

–Bearing carrier face machined, mill operation not included – 1 pt.

–Upright cost as cast but actual part billet machined – 3 pt.

–Pneumatic shifter not included on BOM – 5 pt.

□ The penalties above will be deducted from the points awarded for Accuracy of the Cost Report.

# Rules Overview

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## □ Method B – Adjusted Cost Deductions

$$\text{Penalty} = 2 \times (\text{Table Cost} - \text{Team Reported Cost})$$

- The table cost will be calculated from the standard Cost Tables. The penalty calculation will result in a Rupees value equal to twice the difference between the team cost and the correct cost for all items in error.
- **NOTE:** The table costs of all items in error are included in the calculation. A missing assembly would include the price of all parts, materials, processes and fasteners making up the assembly.

**Applied to Adjusted Cost of vehicle**

- **No penalty** is applied when values are **over estimated**

# Report Structure

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- What should contain in **COST REPORT** :
  - Cover Binder for Cost Report with Name & Car Number on it
  - Table of Contents with page numbers
  - Cost Summary
  - Bill Of Material
  - Individual filled Commodity Section
  
- For report filling **teams have to Refer** Cost tables & Appendix:
  - Table: Materials
  - Table: Process
  - Table: Process Multipliers
  - Table: Fasteners
  - Table: Tooling
  - Appendix C-1 Cost Models and Cost Methodology
  - Appendix C-2 Standard Part Numbering
  - Appendix C-3 Organized List of Systems and Assemblies
  - Appendix C-4 Power Tool Package Envelopes

# Report Structure

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- **Eight Commodities of Cost Report need to be fill :**

*(Sequence has to be as follows in the report)*

|                                       |                     |
|---------------------------------------|---------------------|
| <b>1.Brakes</b>                       | <b>(Code - BR )</b> |
| <b>2.Engine &amp; Drivetrain</b>      | <b>(Code - EN )</b> |
| <b>3.Frame &amp; Body</b>             | <b>(Code - FR )</b> |
| <b>4.Electrical</b>                   | <b>(Code - EL )</b> |
| <b>5.Misc. &amp; Finish</b>           | <b>(Code – MS )</b> |
| <b>6.Steering</b>                     | <b>(Code - ST )</b> |
| <b>7.Suspension</b>                   | <b>(Code – SU )</b> |
| <b>8.Wheels, Bearings &amp; Tires</b> | <b>(Code – WT )</b> |

**(Note : Refer Appendix for parts' list in the commodities)**

# Report Structure

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## □ Decide Make Vs. Buy :

- Made parts are those that start with raw materials (aluminium, etc.), have operations performed on them and may have subassembly
- Bought parts come right off the tables and get directly assembled to vehicle
- If a team genuinely makes a part listed on the table as a bought part they may alternatively cost it as a made part if and only if a place holder entry is listed in the tables enabling them to do so. For example, in the category of dampers a “student built” entry is included. This line item must be included in the BOM (it has zero cost). Then they must proceed to cost the damper they actually designed and built.
- For example a snap ring does not have a “team made” entry in the Cost Tables. A team who made their own would still have to use the table cost based on diameter, even if they could cost it less expensively by buying steel and processing it.
- Any part which is normally purchased that is optionally shown as a made part must have supporting documentation submitted to prove team manufacture. This might include engineering drawings, pictures of machining, etc. **Teams found costing bought parts as made parts will be penalized appropriately.**

# Report Structure

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- **Reference Cost Tables:**
- **Materials Table** -Lists the costs for raw materials used to manufacture parts built by the teams and also of finished parts purchased by the teams.

| Bearings        | Drivetrain  | Safety        |
|-----------------|-------------|---------------|
| Brake System    | Electronics | Sensors       |
| Control Modules | Engine      | Dampers       |
| Fluids          | Tires       | Raw Materials |

- **Raw Materials** – Steel, Aluminium, Titanium
- **Wheels** – BBS, Keizer, JongBloed.
- **Process Table** -Operations which modify materials such as cutting, joining, assembling, etc.

| Material Removal | Joining                | Labor           |
|------------------|------------------------|-----------------|
| Basic Forming    | Laser Cut              | Composites      |
| Tubing           | Electrical Connections | Sheet Materials |

- **Joining** – Sewing, Welding
- **Material Removal** – Turning, Facing, Milling, EDM, Cutting

# Report Structure

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- **Reference Cost Tables: (Continued...)**
- **Process Multipliers -Modify the standard costs of different operations to account for material and geometric differences in the part.**

|                             |
|-----------------------------|
| <b>Assembly</b>             |
| <b>Fastner Installation</b> |
| <b>Drill Tap</b>            |
| <b>Machining</b>            |

- **Fastener Installation – Engagement Length**
- **Drill, Tap – Hole Length**
- **Fasteners - Consists of not only traditional fasteners such as bolts, nuts and rivets but also adhesives, hose clamps and retaining rings**
  - **Fasteners cost by: Grade, Nominal thread size, Length.**
  - **Plumbing fittings are listed in the Materials Table.**

# Report Structure

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- Reference Cost Tables: (Continued...)
- Tooling -The production tooling associated with processes that are specific to the part geometry. E.g. jigs, fixtures, dies, moulds.
  - Tools refer to **hand or power tools** used to assemble the vehicle. The costs of these tools are **not included** in the Cost Report.

|                                   |
|-----------------------------------|
| <b>Die Casting</b>                |
| <b>Lamination</b>                 |
| <b>Plastic injection moulding</b> |
| <b>Powder Metal Forming</b>       |
| <b>Sand Casting</b>               |

- Tooling is process specific and goes only with the processes that require it.
- Column in Process Table labelled “Tooling Required” Will have Yes if needed. Blank means no.
- The costs of Tooling must be included in the Cost Report.

# Report Structure

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## □ Example Cost Table :

- It includes material cost, process cost, fastener cost & tooling costs' subtotal.
- Teams have to fill all data about part name, ID, description and cost.

|          |                          |            |                     |       |    |               |    |       |
|----------|--------------------------|------------|---------------------|-------|----|---------------|----|-------|
| System   | Engine & Drivetrain (EN) | University | University of Akron | Car # | 73 | Part Cost     | \$ | 82.91 |
| Assembly | Fuel Assembly            |            |                     |       |    | Qty           |    | 1     |
| Part     | Fuel Tank                |            |                     |       |    | Extended Cost | \$ | 82.91 |
| P/N Base | P2013                    |            |                     |       |    |               |    |       |
| Suffix   | EN                       |            |                     |       |    |               |    |       |
| Details  | Aluminum Gas Tank        |            |                     |       |    |               |    |       |

| MID | Material                         | Use                | UnitCost | Size1 | Unit1   | Size2 | Unit2 | Area Name          | Area | Length | Density | Quantity  | Sub Total |
|-----|----------------------------------|--------------------|----------|-------|---------|-------|-------|--------------------|------|--------|---------|-----------|-----------|
| 10  | Aluminum, Premium (per kg)       | Fuel Tank Material | \$       | 4.20  | 0.84 kg |       |       | Volume - 1.625 gal |      |        |         | 1         | \$ 3.53   |
| 20  | Fitting, Weld-in, Male, Aluminum | Fuel Line Return   | \$       | 1.72  | 6.35 mm |       |       |                    |      |        |         | 1         | \$ 1.72   |
|     |                                  |                    |          |       |         |       |       |                    |      |        |         |           | \$ -      |
|     |                                  |                    |          |       |         |       |       |                    |      |        |         | Sub Total | \$ 5.25   |

| PID | Process                          | Use                        | UnitCost | Unit      | Quantity | Multiplier | Mult. Val. | Sub Total          |
|-----|----------------------------------|----------------------------|----------|-----------|----------|------------|------------|--------------------|
| 10  | Sheet metal shear                | Cut sheet metal            | \$       | 0.25 cut  |          | 38         |            | \$ 9.50            |
| 20  | Sheet metal bends                | Calculated Bends for Shape | \$       | 0.25 bend |          | 14         |            | \$ 3.50            |
| 30  | Tube end preparation for welding | Prepare for Welding        | \$       | 0.75 end  |          | 38         |            | \$ 28.50           |
| 40  | Weld                             | Weld Gas Tank Seams        | \$       | 0.15 cm   |          | 240        |            | \$ 36.00           |
|     |                                  |                            |          |           |          |            |            | \$ -               |
|     |                                  |                            |          |           |          |            |            | Sub Total \$ 77.50 |

| FID | Fastener                | Use              | UnitCost | Size1 | Unit1 | Size2 | Unit2   | Quantity  | Sub Total |
|-----|-------------------------|------------------|----------|-------|-------|-------|---------|-----------|-----------|
| 10  | Bolt, Grade 8.8 (SAE 5) | Fasten Fuel Tank | \$       | 0.06  | 5 mm  |       | 38 mm   | 2         | \$ 0.12   |
| 20  | Bolt, Grade 8.8 (SAE 5) | Fasten Fuel Tank | \$       | 0.02  | 5 mm  |       | 6.35 mm | 2         | \$ 0.04   |
|     |                         |                  |          |       |       |       |         |           | \$ -      |
|     |                         |                  |          |       |       |       |         | Sub Total | \$ 0.16   |

| TID | Tooling | Use | UnitCost | Unit | Quantity | PVF | FracIncid | Sub Total      |
|-----|---------|-----|----------|------|----------|-----|-----------|----------------|
|     |         |     |          |      |          |     |           | \$ -           |
|     |         |     |          |      |          |     |           | \$ -           |
|     |         |     |          |      |          |     |           | Sub Total \$ - |

# Report Structure

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## □ Bill of Material Structure :

- Cost Report is a costed Bill of Material (BOM)
- Hierarchy: System Cost >>Assembly Cost >>Part Cost >>Material Cost >>Process Cost >>Fastener Cost>> Tooling Cost.
- After filling commodities table of individual systems, fill up following BOM sheet.

| Line Num. | Area of Commodity | Asm # | Asm                  | Part # | Component       | Description                       | Unit Cost | Quantity | Material Cost | Process Cost | Fastener Cost | Tooling Cost | Total Cost  | Details Page Number |
|-----------|-------------------|-------|----------------------|--------|-----------------|-----------------------------------|-----------|----------|---------------|--------------|---------------|--------------|-------------|---------------------|
| 1         | Brake System      | A1001 | Front Rotor Assembly | -      |                 | Complete Front Brake Asm          | \$ 2.50   | 1        | \$ -          | \$ 2.50      | \$ -          | \$ -         | \$ 2.50     | 10                  |
| 2         |                   |       |                      | P1001  | Brake Disks     | Cast Iron Brake Disk              | \$ 5.31   | 2        | \$ 1.05       | \$ 4.26      | \$ -          | \$ -         | \$ 10.62    | 11                  |
| 3         |                   |       |                      | P1002  | Brake Bobbins   | Aluminum Bobbin                   | \$ 3.47   | 12       | \$ 0.04       | \$ 3.29      | \$ 0.14       | \$ -         | \$ 41.61    | 13                  |
| 4         |                   |       |                      | P1003  | Brake Calipers  | Wilwood, PS-1                     | \$ 49.04  | 2        | \$ 46.00      | \$ 2.80      | \$ 0.24       | \$ -         | \$ 98.08    | 15                  |
| 5         |                   |       |                      | P1004  | Brake Pads      | Iron/Steel Pad                    | \$ 2.72   | 4        | \$ 2.46       | \$ 0.26      | \$ -          | \$ -         | \$ 10.87    | 16                  |
| 6         |                   |       |                      | P1005  | Brake Hats      | Aluminum Hat                      | \$ 8.62   | 2        | \$ 1.93       | \$ 6.29      | \$ 0.40       | \$ -         | \$ 17.24    | 17                  |
| 7         |                   | A1002 | Rear Rotor Assembly  | -      |                 | Complete Rear Brake Asm           | \$ -      | 1        | \$ -          | \$ -         | \$ -          | \$ -         | \$ -        | 19                  |
| 8         |                   |       |                      | P1006  | Brake Disks     | Cast Iron Brake Disk              | \$ 3.86   | 2        | \$ 1.05       | \$ 2.81      | \$ -          | \$ -         | \$ 7.72     | 20                  |
| 9         |                   |       |                      | P1007  | Brake Bobbins   | Aluminum Bobbin                   | \$ 3.47   | 12       | \$ 0.04       | \$ 3.29      | \$ 0.14       | \$ -         | \$ 41.61    | 22                  |
| 10        |                   |       |                      | P1008  | Brake Calipers  | Wilwood, PS-1                     | \$ 49.04  | 2        | \$ 46.00      | \$ 2.80      | \$ 0.24       | \$ -         | \$ 98.08    | 24                  |
| 11        |                   |       |                      | P1009  | Brake Pads      | Iron/Steel Pad                    | \$ 2.72   | 4        | \$ 2.46       | \$ 0.26      | \$ -          | \$ -         | \$ 10.87    | 25                  |
| 12        |                   |       |                      | P1010  | Brake Hats      | Aluminum Hat                      | \$ 8.62   | 2        | \$ 1.93       | \$ 6.29      | \$ 0.40       | \$ -         | \$ 17.24    | 26                  |
| 13        |                   | A1003 | Brake Line Assembly  | -      |                 | Complete Brake Line Asm           | \$ 3.25   | 1        | \$ 0.75       | \$ 2.50      | \$ -          | \$ -         | \$ 3.25     | 28                  |
| 14        |                   |       |                      | P1011  | Brake Line      | Steel Braided Lines               | \$ 18.34  | 1        | \$ 16.47      | \$ 1.35      | \$ 0.52       | \$ -         | \$ 18.34    | 29                  |
| 15        |                   |       |                      | P1012  | Fluid Reservoir | Hydraulic Fluid Reservoir         | \$ 16.67  | 1        | \$ 16.54      | \$ 0.13      | \$ -          | \$ -         | \$ 16.67    | 30                  |
| 16        |                   |       |                      | P1013  | Master Cylinder | Master Cylinder, Tilton, Model 77 | \$ 198.62 | 2        | \$ 197.50     | \$ 0.63      | \$ 0.49       | \$ -         | \$ 397.24   | 32                  |
| 17        |                   |       |                      | P1014  | Fittings        | Brake Line Fittings               | \$ 138.32 | 1        | \$ 108.32     | \$ 30.00     | \$ -          | \$ -         | \$ 138.32   | 34                  |
| 18        |                   |       |                      | P1015  | Balance Bar     | Balance Bar, Tilton, 72-281       | \$ 314.63 | 1        | \$ 312.50     | \$ 2.13      | \$ -          | \$ -         | \$ 314.63   | 35                  |
|           | Brake System      |       |                      |        | Area Total      |                                   |           |          | \$ 1,066.18   | \$ 171.30    | \$ 7.42       | \$ -         | \$ 1,244.90 |                     |

# Report Structure

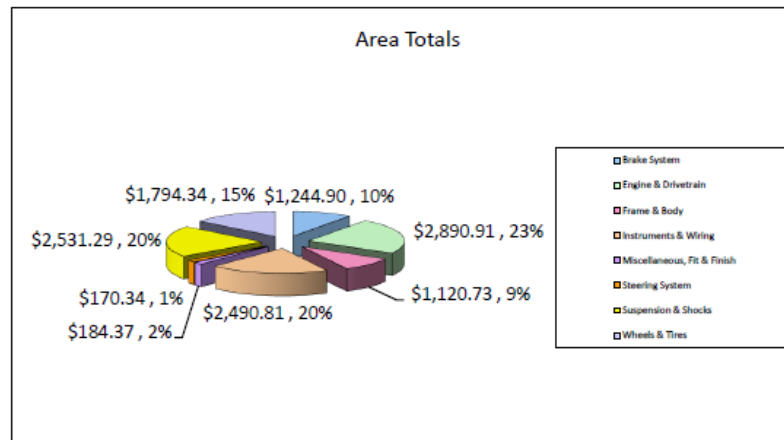
16

## □ Cost Summary:

- Next is to fill up Cost Summary sheet as shown in the fig.
- Start filling Cost Report in reverse manner.

| Area Totals   |                             | Materials   | Processes   | Fasteners | Tooling   | Total        |
|---------------|-----------------------------|-------------|-------------|-----------|-----------|--------------|
| BR            | Brake System                | \$ 1,066.18 | \$ 171.30   | \$ 7.42   | \$ -      | \$ 1,244.90  |
| EN            | Engine & Drivetrain         | \$ 2,347.20 | \$ 506.22   | \$ 37.49  | \$ -      | \$ 2,890.91  |
| FR            | Frame & Body                | \$ 470.83   | \$ 499.32   | \$ 50.66  | \$ 99.93  | \$ 1,120.73  |
| EL            | Instruments & Wiring        | \$ 2,178.52 | \$ 312.30   | \$ -      | \$ -      | \$ 2,490.81  |
| MS            | Miscellaneous, Fit & Finish | \$ 121.24   | \$ 62.51    | \$ 0.62   | \$ -      | \$ 184.37    |
| ST            | Steering System             | \$ 107.30   | \$ 53.52    | \$ 9.50   | \$ 0.02   | \$ 170.34    |
| SU            | Suspension & Shocks         | \$ 1,835.47 | \$ 586.93   | \$ 44.88  | \$ 64.01  | \$ 2,531.29  |
| WT            | Wheels & Tires              | \$ 997.30   | \$ 762.32   | \$ 14.72  | \$ 20.00  | \$ 1,794.34  |
| Total Vehicle |                             | \$ 9,124.03 | \$ 2,954.42 | \$ 165.29 | \$ 183.96 | \$ 12,427.69 |

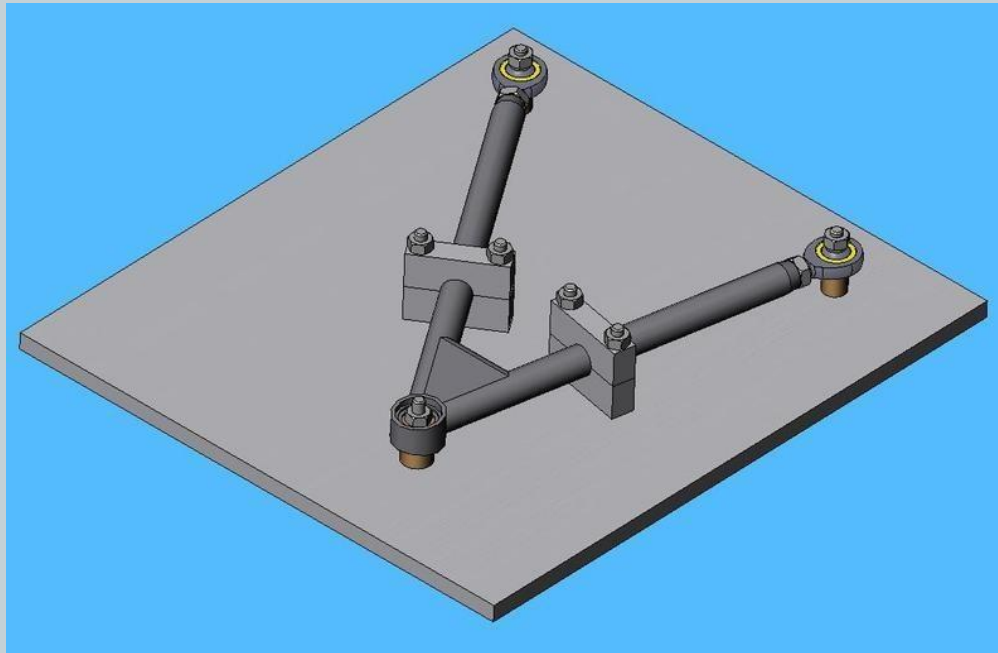
Composition for total vehicle:



# Example

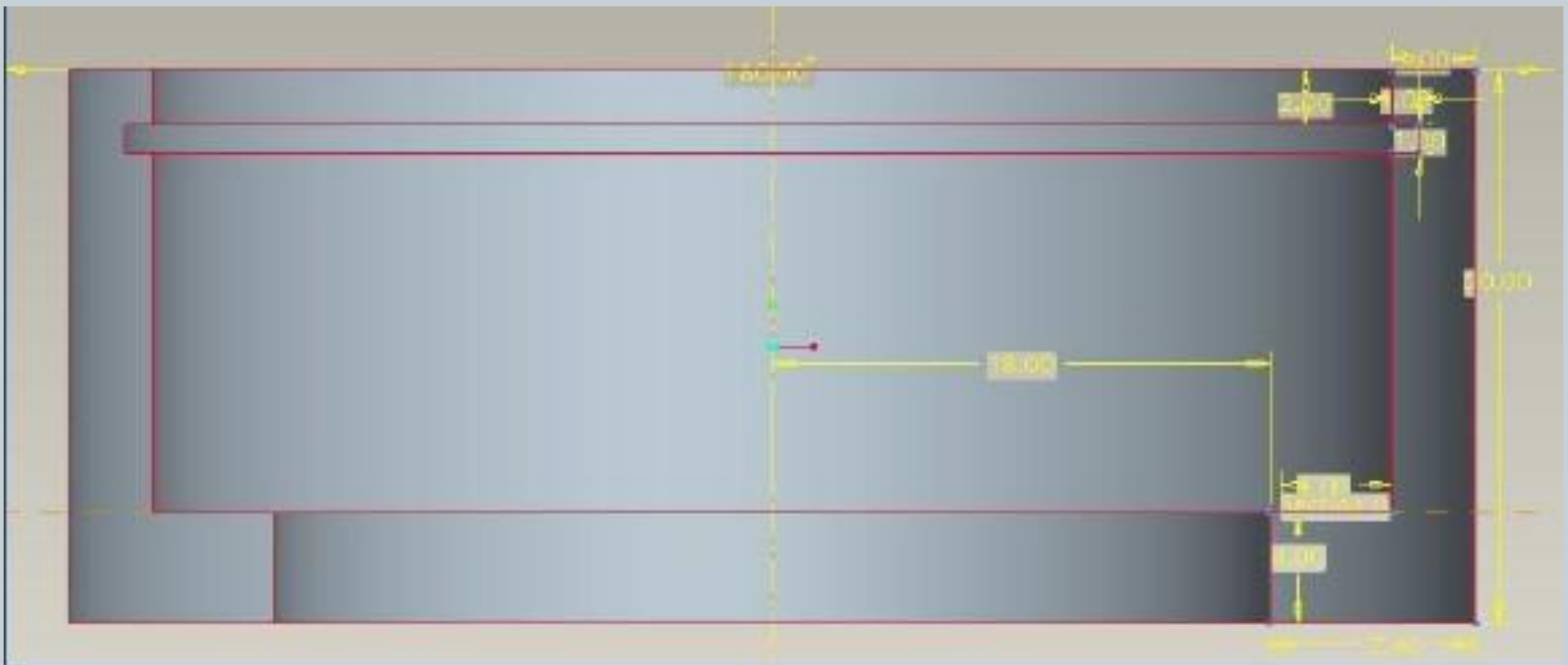
17

- An A-Arm will be used as an example of working with the Cost Tables
- In this illustration, costs are reported in dollars where teams have to submit cost report in Rupees.
- Teams could use this example for reference.



## 18

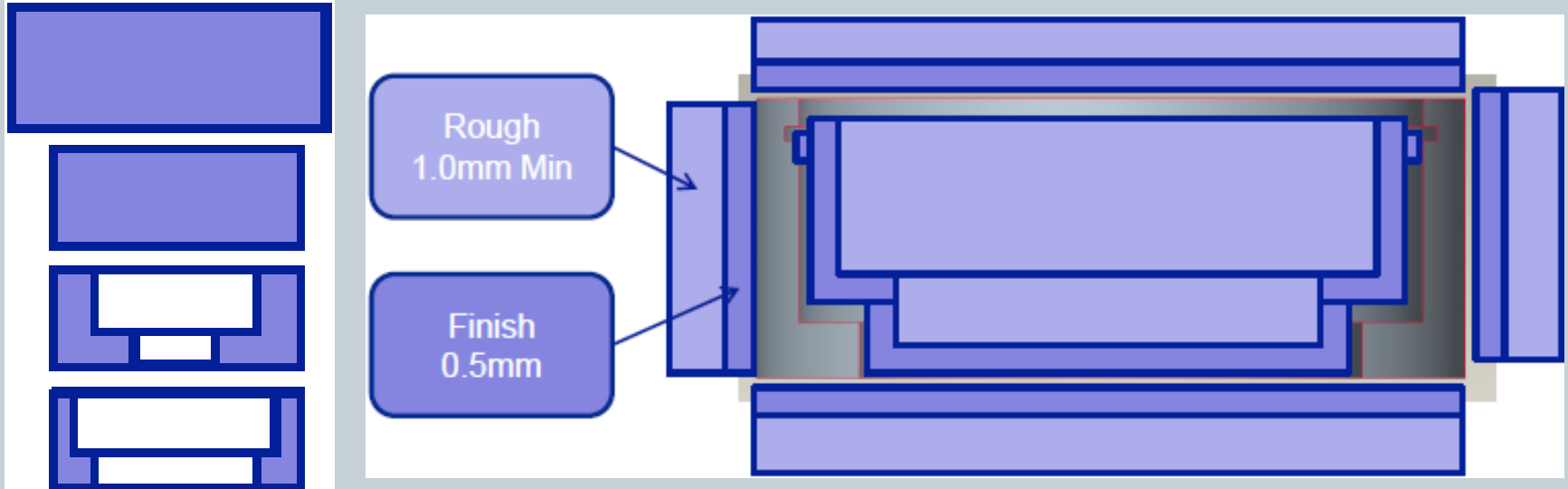
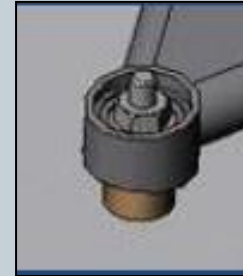
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# Example

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- **Machining :**
- **Process Steps**
  1. Start with steel cylinder
  2. Rough & Finish Machine Outside
  3. Rough Machine Inside
  4. Finish machine Inside



# Example

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## □ How to fill part's cost in the table

- First calculate raw material size before machining keeping rough and finish allowance.
  - Evaluate cross section area, length & density of raw material note that down in material cost block.
  - Refer to materials table for unit cost, then multiply to get material subtotal.
  - In the process cost block, mention all the manufacturing processes done on raw material till it get finished.
  - Write down each description and use in the table.
  - Again refer Process table for unit process costs, units.
  - Now, determine material removed/ Quantity.
  - Further, refer multiplier table for corresponding multiplier value.
  - Mention multiplier name and multiplier value.
  - Calculate Process Subtotal
  - Similarly, mention fasteners, their quantity, unit cost used in the fasteners block with referring Fasteners table.
- e.g. Bearing, ring, washers, bolts, nuts.
- About tooling cost, refer to tooling table for unit cost & calculate PVF, quantity and subtotal. Add all subtotals to get part cost.

# Example

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- ❑ **Sub-Assembly:**
- ❑ **Review your work!**
- ❑ **Did we forget anything?**
- ❑ **Jam nuts were included on BOM but no installation step was included**
- ❑ **The bearing cups and threaded inserts did not include a setup cost in their machining operations**
- ❑ **These would be fed from continuous stock through the machine spindle**
- ❑ **An upright or other part made from billet would need to include a “Machining Setup, Install/Remove” and probably “Machining Setup, Change”**
- ❑ **Include pictures or sketches of your fixtures to avoid discussion at the event**

# Do's & Don'ts

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- ❑ **Do start your BOM NOW!**
  - Distribute the work
- ❑ **Do ask questions early**
- ❑ **When in doubt be conservative**
- ❑ **Don't miss any parts OR processes from your Cost Report**
- ❑ **Don't submit your Cost Report late**
- ❑ **Do prepare for the Real Case.**
- ❑ **Do submit your Cost Addendum Form at the Event Registration**
- ❑ **Don't miss your assigned time at the Event**
- ❑ **Don't forget costs of**
  - Safety wiring
  - Paint
  - Welding jig tooling
  - Tie wraps/Zip ties
  - And many such small small things.....